

Unit 108 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 63.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 31 plots, 344 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (SE 2.6 %)	Type	Sierran mixed conifer
Trees per acre	68	CWHR size / density	5 / D (cover 91 %)
QMD	24.5 in	SDI	274 (47 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	222 sq ft/ac 24.5 in	BA QMD	0 sq ft/ac 0.0 in
TPA	68	TPA	0
Volume	52.3 MBF/ac (9501 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~164 sq ft/ac BA	Sampling error	5.4 % at 95 % (t = 2.04, n = 31)
Species (BA)	WF 68, PP 66, IC 36, DF 29, SP 22	Stratum standard	40 % - MEETS; 31 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	4	3.6	2.6	81	0.4
12-15.9	22	13.2	14.2	425	2.3
16-19.9	33	12.0	21.3	744	4.1
20-23.9	53	12.7	34.2	1310	7.2
24-27.9	60	10.7	38.7	1582	8.7
28-31.9	44	5.9	28.4	1227	6.7
32-35.9	39	4.0	25.2	1152	6.3
36-39.9	35	2.9	22.6	1100	6.1
40-43.9	26	1.7	16.8	860	4.7
44-47.9	6	0.3	3.9	188	1.0
48-51.9	7	0.3	4.5	250	1.4
52-55.9	6	0.2	3.9	230	1.3
56-59.9	5	0.2	3.2	186	1.0
60-63.9	4	0.1	2.6	166	0.9
All	344	68.1	221.9	9501	52.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	106	22.5	68.4	23.6	2863	15.7
PP	103	19.0	66.5	25.3	2891	15.9
IC	56	12.9	36.1	22.7	1471	8.1
DF	45	7.0	29.0	27.6	1290	7.1
SP	34	6.7	21.9	24.4	986	5.4
All	344	68.1	221.9	24.5	9501	52.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 108: plots on the 300 ft grid (orange = QA flag)

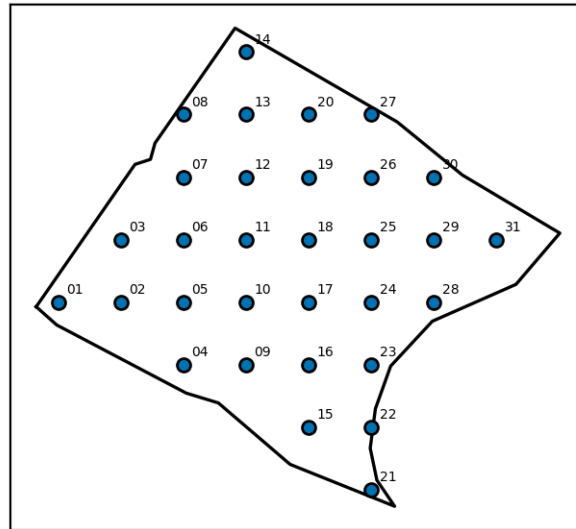


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
108-01	10	200	11	ok
108-02	11	220	22	ok
108-03	9	180	23	ok
108-04	10	200	18	ok
108-05	10	200	16	ok
108-06	12	240	21	ok
108-07	11	220	27	ok
108-08	11	220	23	ok
108-09	11	220	27	ok
108-10	12	240	12	ok
108-11	11	220	14	ok
108-12	14	280	25	ok
108-13	6	120	17	ok
108-14	13	260	6	ok
108-15	14	280	15	ok
108-16	10	200	13	ok
108-17	11	220	9	ok
108-18	11	220	8	ok
108-19	9	180	32	ok
108-20	12	240	17	ok
108-21	10	200	22	ok
108-22	12	240	30	ok
108-23	11	220	18	ok
108-24	12	240	11	ok
108-25	10	200	16	ok
108-26	13	260	15	ok
108-27	13	260	19	ok
108-28	11	220	10	ok
108-29	11	220	13	ok
108-30	10	200	20	ok
108-31	13	260	10	ok